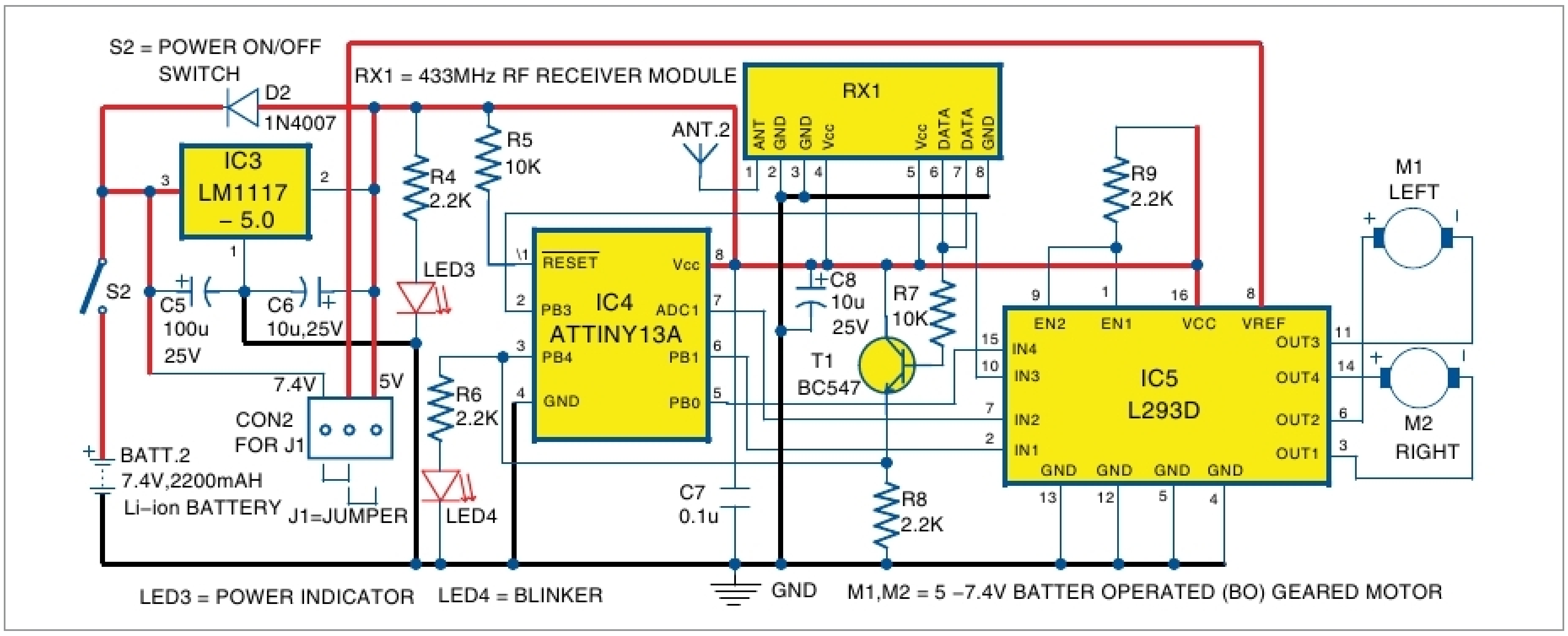

Fig. 4: Circuit diagram of the receiver

to transmit the code, while loop in JoystickRx.C continuously monitors the signal and then sends the proper four-bit data to IC L293D to control the motors.

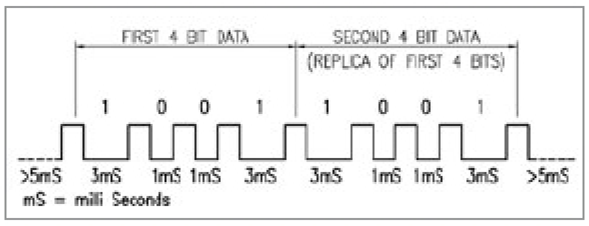

Fig. 5: Timing pulse of eight-bit serial data

Accordingly, JoystickTx.hex is generated to burn into ATtiny13A (IC2) and JoystickRx.hex to ATtiny13A (IC4). For burning the codes, you can use any suitable ATtiny13A programmer.

Construction and testing

An actual-size PCB layout of the transmitter section is shown in Fig. 6 and its components layout in Fig. 7. Assemble the circuit on the PCB. Connect the joystick module across connector CON1 using external jumper wires. Power supply for the transmitter is provided by 9V battery.

An actual-size PCB layout of the receiver section is shown in Fig. 8 and its components layout in Fig. 9. Make all jumper connections using external wires. Assemble the circuit on the PCB. Power supply for the receiver is provided by 7.4V lithium battery.

After assembling the circuits on transmitter and receiver PCBs without RF modules, burn/write JoystickTx.hex and JoystickRx.hex on the respective ATtiny13A MCUs using a suitable AVR programmer. Then, mount the MCUs on the respective PCBs.

Connect signal output pin 6 of ATtiny13 of the transmitter board and signal input pin 3 of ATtiny13 of the receiver

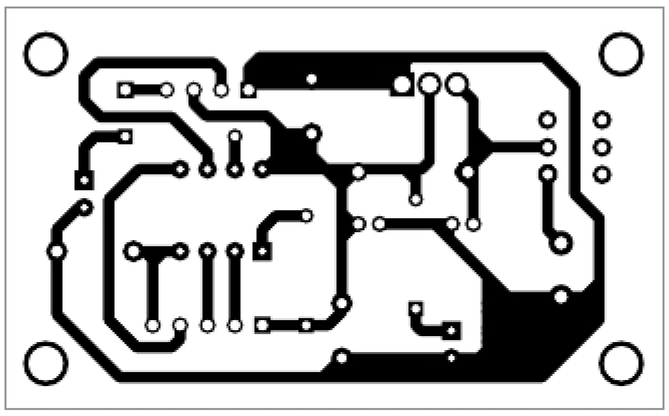

Fig. 6: Actual-size PCB layout of the transmitter unit

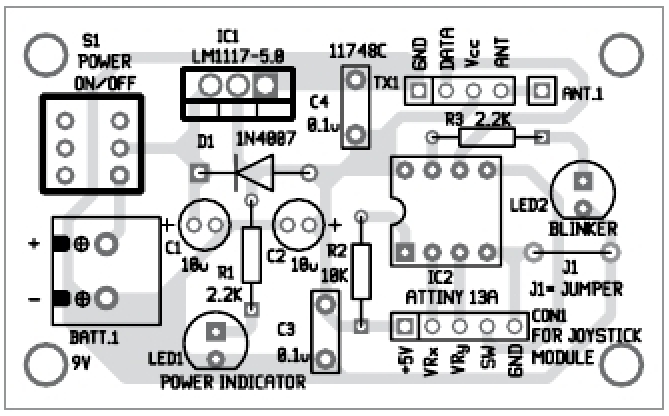

Fig. 7: Components layout for the transmitter PCB